

Create RestTemplate

1-Spring initial



Project

☐ Gradle - Groovy ☐ Gradle - Kotlin ☒ **Java** ☐ Kotlin ☐ Groovy

☒ **Maven**

Spring Boot

☐ 4.2.0 (SNAPSHOT) ☐ 4.2.0 (M2) ☐ 4.1.2 (SNAPSHOT) ☒ **4.1.1**

☐ 4.0.9 (SNAPSHOT) ☐ 4.0.8

Project Metadata

Group

com.vibol

Artifact

RestTemplateDemo

Package name

com.vibol.RestTemplateDemo

Packaging

☒ **Jar** ☐ War

Configuration

☒ **Properties** ☐ YAML

Java

☐ 27 ☒ **25** ☐ 21 ☐ 17

Dependencies

Spring Web **WEB**

Build web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container.

ADD DEPENDENCIES... CTRL + B

2-Add Bean to the application

```
1 package com.vibol.RestTemplateDemo;
2
3 import org.springframework.boot.SpringApplication;
4 import org.springframework.boot.autoconfigure.SpringBootApplication;
5 import org.springframework.context.annotation.Bean;
6 import org.springframework.web.client.RestTemplate;
7
8 @SpringBootApplication
9 public class RestTemplateDemoApplication {
10
11     public static void main(String[] args) {
12         SpringApplication.run(RestTemplateDemoApplication.class, args);
13     }
14
15     // 1. Define the RestTemplate bean here so Spring can manage it
16     @Bean
17     public RestTemplate restTemplate() {
18         return new RestTemplate();
19     }
20 }
```

3- Create Service

```
RestTemplateDemo ▶ src/main/java ▶ com.vibol.RestTemplateDemo ▶ MyService ▶ fetchData(Long) : String
1 package com.vibol.RestTemplateDemo;
2
3 import org.springframework.stereotype.Service;
4 import org.springframework.web.client.RestTemplate;
5
6 @Service
7 public class MyService {
8
9     private final RestTemplate restTemplate;
10
11     public MyService(RestTemplate restTemplate) {
12         this.restTemplate = restTemplate;
13     }
14
15     public String fetchData(Long id) {
16         String url = "https://jsonplaceholder.typicode.com/todos/"+id;
17         return restTemplate.getForObject(url, String.class);
18     }
19
20     public MyRespond fetchDataWithRespon(Long id) {
21         String url = "https://jsonplaceholder.typicode.com/todos/"+id;
22         return restTemplate.getForObject(url, MyRespond.class);
23     }
24 }
```

Package Explorer | Project Explorer

Udemy-Course [udemy-microservice Section12_SetupDB]

- configserver
- order
- product
- RestTemplateDemo [boot] [udemy-microservice Section12_SetupDB]
 - src/main/java
 - com.vibol.RestTemplateDemo
 - MyController.java
 - MyRespond.java
 - MyService.java
 - RestTemplateDemoApplication.java
 - src/main/resources
 - src/test/java
 - IRF System Library [jdk-751]

RestTemplateDemo/pom.xml | RestTemplateDemoApplication.java | MyService.java | MyRespond.java

```
1 package com.vibol.RestTemplateDemo;
2
3 public record MyRespond(Long id, String title) {
4
5 }
6
```

4-Controller

```
RestTemplateDemo/pom.xml  RestTemplateDemoApplication.java  MyService.java  MyController.java  MyRespond.java
RestTemplateDemo > src/main/java > com.vibol.RestTemplateDemo > MyController
1 package com.vibol.RestTemplateDemo;
2
3 import org.springframework.web.bind.annotation.GetMapping;
4 import org.springframework.web.bind.annotation.PathVariable;
5 import org.springframework.web.bind.annotation.RestController;
6
7 @RestController
8 public class MyController {
9
10     private final MyService myService;
11
12     // Constructor injection for the service
13     public MyController(MyService myService) {
14         this.myService = myService;
15     }
16
17     @GetMapping("/get-external-data/{id}")
18     public String getExternalData(@PathVariable Long id) {
19         // Calls the service method that uses RestTemplate
20         return myService.fetchData(id);
21     }
22
23     @GetMapping("/get-external-data1/{id}")
24     public MyRespond getExternalDataWithRespond(@PathVariable Long id) {
25         // Calls the service method that uses RestTemplate
26         return myService.fetchDataWithRespond(id);
27     }
28 }
```

Udemy_Microservice > RestTemplate

GET

Docs Params Authorization Headers 8 Body Scripts Settings

Query Params

Key	Value
Key	Value

Body Cookies Headers 5 Test Results

{ } JSON Preview Visualize

```
1 {
2   "id": 3,
3   "title": "fugiat veniam minus"
4 }
```